

Equivalent Ratios (Sheet 1)

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Multiply these ratios by the given number to create equivalent ratios.

1 $3 : 4 = \square : \square$ $\times 5$

2 $2 : 7 = \square : \square$ $\times 3$

3 $5 : 1 = \square : \square$ $\times 6$

4 $15 : 4 = \square : \square$ $\times 2$

5 $3 : 8 : 2 = \square : \square : \square$ $\times 3$

6 $1 : 7 : 4 = \square : \square : \square$ $\times 6$

Find the value in the top circle, write it in the lower circle as well, then calculate the box.

7 $1 : 2 = 2 : \square$

8 $4 : 3 = 8 : \square$

9 $3 : 7 = 9 : \square$

10 $10 : 3 = 50 : \square$

Now the lower circle first.

11 $14 : 9 = \square : 27$

12 $8 : 5 = \square : 25$

13 $1 : 6 = \square : 42$

14 $10 : 9 = \square : 81$

Find the box values that make equivalent ratios.

15 $4 : 7 = 12 : \square$

16 $3 : 5 = \square : 30$

17 $8 : 1 = \square : 7$

18 $6 : 5 = 36 : \square$

19 $9 : 100 = \square : 1\,000$

20 $20 : 3 = 60 : \square$

21 $13 : 4 = 52 : \square$

22 $11 : 12 = \square : 60$

23 $15 : 4 = 165 : \square$

Instead of boxes, its algebra. There are also some triples.

24 $4 : 9 = x : 36$ $x =$

25 $13 : 7 = y : 21$ $y =$

26 $1 : 19 = 5 : j$ $j =$

27 $20 : 13 = d : 65$ $d =$

28 $5 : 2 : 10 = 20 : a : b$

$a =$ $b =$

29 $3 : 7 : 4 = k : 21 : h$

$k =$ $h =$

30 $5 : 9 : 1 = e : t : 8$

$e =$ $t =$

31 $2 : 7 : 7 = 14 : x : y$

$x =$ $y =$

32 $20 : 3 : 14 = b : a : 126$

$b =$ $a =$

33 $100 : 1 : 10 = w : 10 : x$

$w =$ $x =$

Now use division, because the RHS ratios are smaller than the LHS ratios. Circles then box.

34 $12 : 6 = 6 : \square$

35 $24 : 32 = 6 : \square$

36 $64 : 16 = \square : 2$

37 $27 : 9 = \square : 3$

38 $36 : 18 = 4 : \square$

39 $100 : 125 = 20 : \square$

40 $60 : 48 = \square : 12$

41 $81 : 54 = \square : 2$

42 $144 : 64 = 18 : \square$

Now division & multiplication required. Find the unknowns for these ratios to be equivalent

43 $5 : 3 : x = 15 : b : 12$

$x =$ $b =$

44 $4 : 1 : a = 24 : y : 42$

$a =$ $y =$

45 $36 : c : 54 = q : 12 : 6$

$c =$ $q =$

46 $2 : d : e = 4 : e : 20$

$d =$ $e =$

47 $a : b : 12 = 100 : a : 60$

$a =$ $b =$

$a =$ $b =$

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Multiply these ratios by the given number to create equivalent ratios.

1 $3 : 4 = 15 : 20$

2 $2 : 7 = 6 : 21$

3 $5 : 1 = 30 : 6$

4 $15 : 4 = 30 : 8$

5 $3 : 8 : 2 = 9 : 24 : 6$

6 $1 : 7 : 4 = 6 : 42 : 24$

Find the value in the top circle, write it in the lower circle as well, then calculate the box.

7 $1 : 2 = 2 : 4$

8 $4 : 3 = 8 : 6$

9 $3 : 7 = 9 : 21$

10 $10 : 3 = 50 : 15$

Now the lower circle first.

11 $14 : 9 = 42 : 27$

12 $8 : 5 = 40 : 25$

13 $1 : 6 = 7 : 42$

14 $10 : 9 = 90 : 81$

Find the box values that make equivalent ratios.

15 $4 : 7 = 12 : 21$

16 $3 : 5 = 18 : 30$

17 $8 : 1 = 56 : 7$

18 $6 : 5 = 36 : 30$

19 $9 : 100 = 90 : 1\,000$

20 $20 : 3 = 60 : 9$

21 $13 : 4 = 52 : 16$

22 $11 : 12 = 55 : 60$

23 $15 : 4 = 165 : 44$

Instead of boxes, its algebra. There are also some triples.

24 $4 : 9 = x : 36$ $x = 16$

25 $13 : 7 = y : 21$ $y = 39$

26 $1 : 19 = 5 : j$ $j = 95$

27 $20 : 13 = d : 65$ $d = 100$

28 $5 : 2 : 10 = 20 : a : b$

$a = 8$ $b = 40$

29 $3 : 7 : 4 = k : 21 : h$

$k = 9$ $h = 12$

30 $5 : 9 : 1 = e : t : 8$

$e = 40$ $t = 72$

31 $2 : 7 : 7 = 14 : x : y$

$x = 49$ $y = 49$

32 $20 : 3 : 14 = b : a : 126$

$b = 180$ $a = 27$

33 $100 : 1 : 10 = w : 10 : x$

$w = 1\,000$ $x = 100$

Now use division, because the RHS ratios are smaller than the LHS ratios. Circles then box.

34 $12 : 6 = 6 : 3$

35 $24 : 32 = 6 : 8$

36 $64 : 16 = \square : 2$

37 $27 : 9 = 9 : 3$

38 $36 : 18 = 4 : 2$

39 $100 : 125 = 20 : 25$

40 $60 : 48 = 15 : 12$

41 $81 : 54 = 3 : 2$

42 $144 : 64 = 18 : 8$

Now division & multiplication required. Find the unknowns for these ratios to be equivalent

43 $5 : 3 : x = 15 : b : 12$

$x = 4$ $b = 9$

44 $4 : 1 : a = 24 : y : 42$

$a = 7$ $y = 6$

45 $36 : c : 54 = q : 12 : 6$

$c = 108$ $q = 4$

46 $2 : d : e = 4 : e : 20$

$e = 10$ $d = 5$

47 $a : b : 12 = 100 : a : 60$

$a = 20$ $b = 4$

Equivalent Ratios (Sheet 2)

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Find the missing values.

1 $3 : \square = 12 : 24$

2 $\square : 15 = 2 : 5$

3 $8 : 28 = 42 : \square$

4 $4 : \square = 18 : 54$

5 $\square : 20 = 9 : 6$

6 $6 : 8 = \square : 84$

7 $\square : 3 = 172 : 43$

8 $5 : \square = 36 : 144$

9 $12 : 8 : 4 = 39 : a : b$

$a =$ $b =$

10 $3 : 12 : n = q : 78 : 26$

$n =$ $q =$

11 $16 : x : 4 = e : 225 : 45$

$e =$ $x =$

12 $b : 45 : b = d : 6 : 100$

$b =$ $d =$

Use equivalent ratios to solve these.

13 Sam's birthday balloons are coloured silver and gold in the ratio 3 : 5. If there are 27 silver balloons, how many gold balloons are there?

$\square : \square = \square : \square$

Write your answer in words

14 How many balloons were there altogether?

15 Two-stroke fuel for a lawn mower requires an oil : petrol ratio of 1 : 40. How much petrol is added to 50 mL of oil?

$\square : \square = \square : \square$

Write your answer showing units

16 How much fuel is there? Give your answer in litres.

17 Sandra is an experienced electrician. She can install 7 light fittings while her apprentice fits 2 in the same time. If Sandra installs 125 lights, how many has the apprentice installed?

$\square : \square = \square : \square$

18 A street directory has a scale of 1 : 20 000. If a street is 9.2 cm long in the directory, give its actual length in metres.

19 If a street was 720 m in length, how long would the street be in the directory. Give your answer in cm.

Rangers use 'Capture & Release' techniques and ratios to calculate animal numbers. Use tag results to estimate animal populations.

20 Wildlife rangers catch 48 deer, tag them and release them. Two months later 29 deer are caught of which 3 have tags. Estimate the park's deer population, 'n'.

1st Capture: 2nd Capture:

$48 : n = 3 : 29$
Tag Pop Tag Pop

21 Lake Scaley rangers catch 100 trout, tag them, then release them back into the lake. Three weeks later they catch 55 trout of which 4 are tagged. Estimate the number of trout in the lake.

1st Capture: 2nd Capture:

$\square : \square = \square : \square$
Tag Pop Tag Pop

22 Mount Bugs National Park rangers are monitoring rabbits. They catch 205, tag and release them. Later they catch 320 rabbits of which 8 are tagged. Estimate the rabbit population.

1st Capture: 2nd Capture:

$\square : \square = \square : \square$
Tag Pop Tag Pop

Equivalent Ratios (Sheet 2)

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Find the missing values.

1 $3 : \boxed{6} = 12 : 24$

2 $\boxed{6} : 15 = 2 : 5$

3 $8 : 28 = 42 : \boxed{147}$

4 $4 : \boxed{12} = 18 : 54$

5 $\boxed{30} : 20 = 9 : 6$

6 $6 : 8 = \boxed{63} : 84$

7 $\boxed{12} : 3 = 172 : 43$

8 $5 : \boxed{20} = 36 : 144$

9 $12 : 8 : 4 = 39 : a : b$

$a = 26$ $b = 13$

10 $3 : 12 : n = q : 78 : 26$

$n = 4$ $q = 19.5$

11 $16 : x : 4 = e : 225 : 45$

$e = 180$ $x = 20$

12 $b : 45 : b = d : 6 : 100$

$b = 750$ $d = 100$

Use equivalent ratios to solve these.

13 Sam's birthday balloons are coloured silver and gold in the ratio 3 : 5. If there are 27 silver balloons, how many gold balloons are there?

$\boxed{3} : \boxed{5} = \boxed{27} : \boxed{g}$

There are 45 gold balloons.

Write your answer in words

14 How many balloons were there altogether?

$27 + 45 = 72$

There are 72 balloons.

15 Two-stroke fuel for a lawn mower requires an oil : petrol ratio of 1 : 40. How much petrol is added to 50 mL of oil?

$\boxed{1} : \boxed{40} = \boxed{50} : \boxed{p}$

2 000 ml of petrol is added.

Write your answer showing units

16 How much fuel is there? Give your answer in litres.

$2\ 000 + 50 = 2\ 050\text{ mL}$

There is 2.05 L of fuel.

17 Sandra is an experienced electrician. She can install 7 light fittings while her apprentice fits 2 in the same time. If Sandra installs 125 lights, how many has the apprentice installed?

$\boxed{7} : \boxed{2} = \boxed{125} : \boxed{a}$

Apprentice has fitted 35.7 (1 dp) lights, so 35 lights installed.

18 A street directory has a scale of 1 : 20 000. If a street is 9.2 cm long in the directory, give its actual length in metres.

$1 : 20\ 000 = 9.2 : l$

$\text{Length} = 184\ 000\text{ cm}$

$\text{Length of street is } 1\ 840\text{ m.}$

19 If a street was 720 m in length, how long would the street be in the directory. Give your answer in cm.

$1 : 20\ 000 = l : 720$

$\text{Length} = 0.036\text{ m}$

$\text{Length in book is } 3.6\text{ cm.}$

Rangers use 'Capture & Release' techniques and ratios to calculate animal numbers. Use tag results to estimate animal populations.

20 Wildlife rangers catch 48 deer, tag them and release them. Two months later 29 deer are caught of which 3 have tags. Estimate the park's deer population, 'n'.

1st Capture: 2nd Capture:

$\boxed{48} : \boxed{n} = \boxed{3} : \boxed{29}$

Tag Pop Tag Pop

$16 \times 29 = 464\text{ deer.}$

There are approximately 464 deer in the park.

21 Lake Scaley rangers catch 100 trout, tag them, then release them back into the lake. Three weeks later they catch 55 trout of which 4 are tagged. Estimate the number of trout in the lake.

1st Capture: 2nd Capture:

$\boxed{100} : \boxed{n} = \boxed{4} : \boxed{55}$

Tag Pop Tag Pop

$25 \times 55 = 1\ 375\text{ trout.}$

There are approximately 1 375 trout in the lake.

22 Mount Bugs National Park rangers are monitoring rabbits. They catch 205, tag and release them. Later they catch 320 rabbits of which 8 are tagged. Estimate the rabbit population.

1st Capture: 2nd Capture:

$\boxed{205} : \boxed{n} = \boxed{8} : \boxed{320}$

Tag Pop Tag Pop

$25.625 \times 320 = 8\ 200\text{ rabbits.}$

There are approximately 8 200 rabbits in the park.